DoseBot- Smart Pill Dispenser with Quality Verification and Community Healthcare Access

BSc (Hons) in Engineering, University of Moratuwa, semester 04, EE 2044 project - Group 29

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1. Introduction

We, as semester 04 engineering undergraduates, are going to develop a LabVIEW-controlled medical healthcare assistance and quality maintenance smart system named "DoseBot." This Smart IoT system will be consist of mainly sensors actuators which are driving mainly on LABVIEW software and Data Acquisition(DAQ) cards So that the group will have a broader knowledge regarding the planning, coding, debugging and implementation skills. A python based Computer vision system and interactive Web portal will be the other main implementations in this kiosk creating this a multi-modal smart IoT system. We have an idea to pitch this to community services, offices, elderly homes, and for public places so that patients can directly or indirectly contacting the doctor and take the needed medication right through our kiosk without wasting any time on appointments and long queues.

1.1 Importance of the system

In rural areas or in crowded areas, the convenience of proper medication is a challenge. In rural areas public maternal clinics and community services exist, but the problem is the lack of doctors. So that the proper dosages or even for small headaches, people are having difficulty in getting proper medication. But our system implement in community centers will be reducing this problem. Even also in crowded places these types of kiosks, will be providing a huge service, limiting the crowd.

1.2 Objectives

- Hardware integration - Integrating the system using NI DAQ card and LabVIEW software, getting sensor outputs and controlling the actuators; There are 4 or 5 compartments with commonly used medications with proper medicine storage, and the servo starts and releases it by a controlled signal.
- Custom sensor design - We are planning to develop a custom built bottle presence sensor using the strain gauge mechanism.
- Visual verification - We are going to implement computer vision using Python, integrating a webcam so that if the patient got the verified pills and number of pills are correct.
- Web interface - This will be the communication mode of the kiosk so that it have two modes connecting the doctor, and he will recommend the medication or the person can get the pills. We are here going to use a Sinhala voice chatbot service, as it will be very helpful in realtime conversation.

2. Background and Context

2.1 Origin of the Concept: From Healthcare Delays to Automation

The concept for this project originated from a practical inefficiency observed within the public healthcare system. A member of our group visited a government rural hospital for a minor illness and, after several hours of waiting, was prescribed only basic medication that could have been obtained directly from a pharmacy.

This experience highlighted the unnecessary burden placed on hospitals for non-critical cases, leading to long waiting times and inefficient use of medical resources. In many instances, once a diagnosis is established, the process of obtaining medication remains centralized and time-consuming. This observation led to the idea of developing an automated pill dispensing system that operates post-diagnosis, reducing hospital congestion while improving access to essential medication. Such systems align with global efforts to address inefficiencies in medication access and adherence [1].

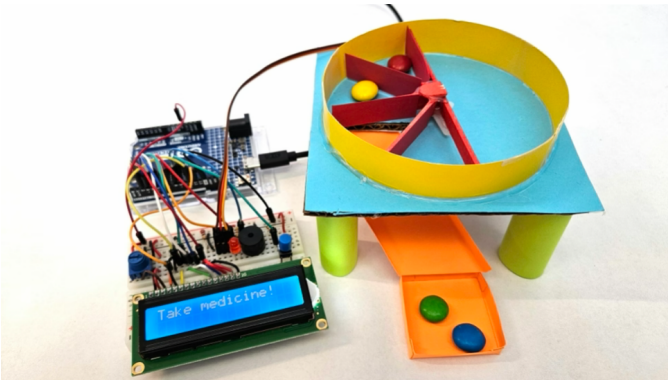

Figure 1: Pill Dispenser using arduino logics

2.2 Technological Landscape: Evolution of Hardware and Mechanical Design

Automated dispensing systems have evolved from passive storage units to active, programmable devices integrating mechanical actuation and embedded control systems. These systems aim to minimize human error while ensuring timely and accurate medication delivery [2]. Different mechanical architectures have been explored in existing implementations, each offering trade-offs between simplicity, scalability and reliability.

Feature	Rotating Disk Mechanism	Gear-Based Stepper
Actuator Type	Two-phase motor (L293D)	28BYJ-48 stepper (ULN2003)
Dispensing Logic	Single-hole alignment	Gear-driven multi-compartment
Scalability	Limited (3 compartments)	Higher (up to 15 compartments)
Application	Basic prototype systems	Residential smart dispensers

Table 1: Comparison of mechanical architectures

The rotating disk mechanism offers simplicity and ease of control but is limited in compartment capacity, making it suitable for basic applications. In contrast, gear-driven stepper systems enable higher compartment density and improved scheduling flexibility, supporting more complex medication regimens. More advanced systems extend this capability to over 20 compartments, enabling long-term medication management [3].

Figure 2: Another prototype pill dispenser - Source instructible.com

Beyond mechanical design, the control architecture plays a crucial role in system reliability. Most academic prototypes utilize microcontrollers such as Arduino or ESP-based platforms due to their low cost and accessibility. However, these systems may lack deterministic timing and robust signal conditioning required for safety-critical applications. Industrial-grade approaches incorporate structured control environments and dedicated data acquisition systems, ensuring precise synchronization between sensing and actuation. This enables real-time monitoring, improved fault detection and enhanced system stability [4].

2.3 Strategic Gap and Proposed Direction

Despite advancements in automated dispensing technologies, a gap remains between expensive hospital-grade systems and low-cost consumer devices with limited reliability. Accessibility and affordability continue to be major challenges, particularly in resource-constrained environments [5].

Figure 3: Modern Pill dispenser kiosk

The proposed system addresses this gap through a community-based kiosk model, enabling shared access to a reliable dispensing platform while reducing individual cost. Additional features such as environmental monitoring and weight-based verification enhance system reliability and ensure proper medication handling. Overall, this approach supports the shift toward decentralized healthcare by integrating automation with practical deployment strategies, offering a scalable and cost-effective solution for community-level medication management.

3. Project Scope

In-scope:

- Using recycled material, we are developing the delivery bay and protective outer case of the kiosk.
- NI DAQ card integration
- Sensor integration: Mainly our idea is to use humidity/temperature sensor development for medicine store part with integration of cooling fan and warning system, then integrate CV camera sensor and a strain gauge sensor build from ground to up using our unique method and a sensor to identify the count of pills (IR sensor).
- Actuators will be servo motors for dispense wheels control and cooling fan.
- Python and Simulink pipeline for CV.
- Web interface with using AWS services and APIs for chatbot, cloud data storage and communication.

Out of scope:

- Building web applications, backup power development, security measures for malicious and vulnerabilities, reusability and remote monitoring.

4. Methodology

In order to divide the workload among the members equally, we divide the subsystem into three main parts.

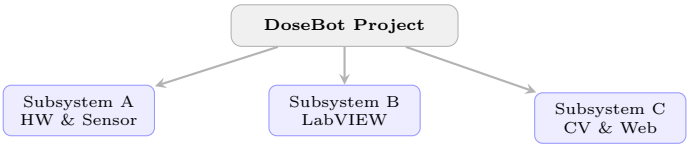

Figure 4: Team workload split across three main subsystems

4.1 1. Hardware and Sensor design

The member should have prominent knowledge in planning, developing, and making the practical design with the use of efficient material and time management. He will be mainly looking into our 3 main sensors temperature and humidity, strain gauge sensor, and IR reflectance sensor.

As strain gauge sensors have small amplitude signals, we should develop a good amplification system to amplify the system, and we have an idea to use a Wheatstone bridge with Opamps or LabVIEW integration (like an ultrasonic sensor).

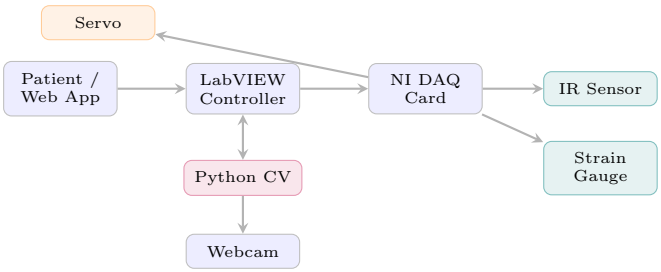

Figure 5: System architecture overview

4.2 2. LabVIEW DAQ Control Logic

LabVIEW will act as the master controller. The LabVIEW program interfaces with the DAQ card to collect sensor data, analyze and send signals to the actuators. Figure 6 shows the full dispensing logic.

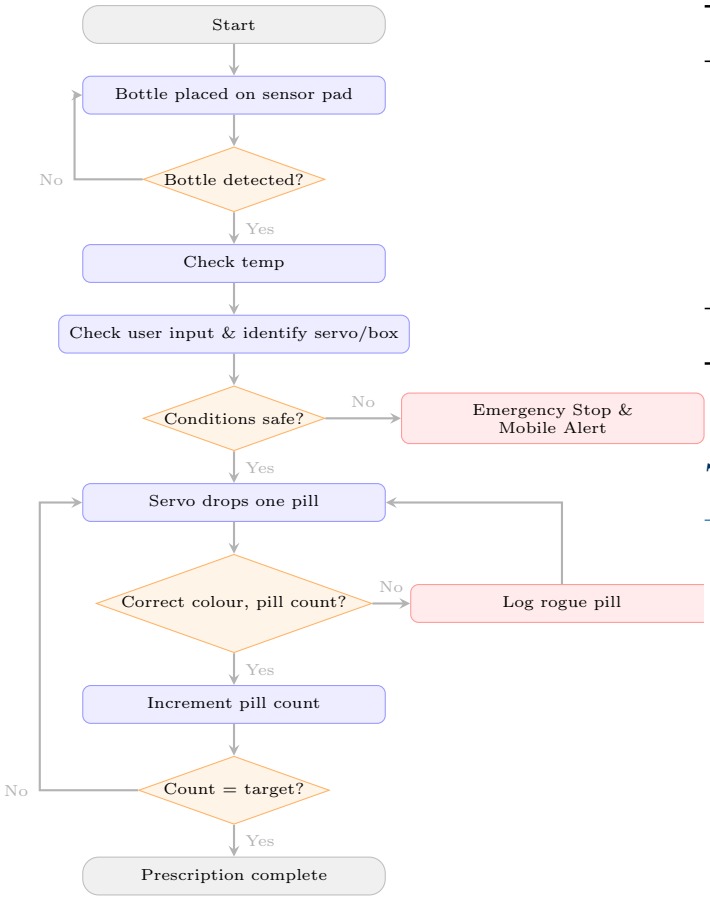

Figure 6: Dispensing operation flowchart

4.3 3. Computer Vision and Web Interface

Webcam and CV detect the pills and environment and any anomaly there. Then using programming and labview integration of the system we are trying to A QR-code-accessible HTML/JS web application allows patients to initiate a dispense cycle from any smartphone browser. We are trying to implement using AWS Bedrock and Lex for our Gen AI integration and chatbot.

5. Timeline

Wk	Milestone	Key Activities
1	Design & Procurement	CAD sketches; procure DAQ, sensors, servo
2	Custom Sensor	Build Wheatstone bridge; calibrate circuit
3	LabVIEW Integration	Wire components; implement logic
4	CV & Web App	OpenCV HSV script; QR web application
5	Assembly & Testing	Full build; accuracy tests; demo preparation

Table 2: Five-week project timeline

6. Estimated Budget

Item	Cost (Rs)
wooden planks/ delivery bay production	1500
servo motors (400*4)	1600
plastic containers	1000
medicine funnels	1000
IR sensor	100
other = wires and connectors ins	1500
Total	**6700**

Table 3: Estimated prototype budget

7. Further Discussion

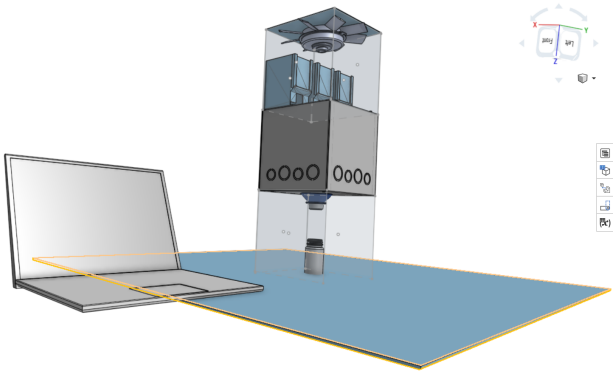

Figure 7: preliminary Design of system using Solidworks

- The system needs a fully CAD model and then design the sensor stages so that we are going over the internet and collecting the data about the sensor integration and developing and calibration of the sensors.
- These bio medical devices are very sensitive so we need extra care for the medicinal security of the system, so initially we contacted the divisional MOH office , Imaduwa (Galle, Sri Lanka), and get their idea about how to store and how to carefully use the medicines, and we are going according to their valuable instructions.
- This product will be eco-friendly and trying to minimize the wastage as well as use of good practices of materials and disposal of them,

8. Conclusion

Our DoseBot addresses a critical gap in rural health care as well as medication queues by merging LabVIEW/NI DAQ control with computer vision and new web technologies. We are thinking

that this will be a unique implementation for Sri Lanka's telemedicine. Also we would like to convey thanking for the guidance of Dr. Logeesan of the Electrical Engineering Department at University of Moratuwa for his valuable insights for the project and the valuable insights from the public health officers in MOH office Imaduwa for their valuable insights regarding the project .

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